

TrustLens: An End-to-End Framework for Fairness Auditing, Bias Mitigation, and Explainable AI

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1. Introduction

The rapid adoption of machine learning systems in domains such as hiring, finance, healthcare, and education has raised important concerns regarding fairness and accountability. While predictive models can improve efficiency and decision-making, they may also inherit and amplify biases present in historical data. Such biases can lead to unfair outcomes for individuals belonging to protected demographic groups, resulting in ethical, social, and legal challenges. Consequently, ensuring fairness in machine learning systems has become a critical area of research within the broader field of Responsible Artificial Intelligence (AI).

TrustLens is an explainable fairness auditing framework designed to evaluate, analyze, and mitigate demographic bias in machine learning models. The project provides an end-to-end workflow that enables users to inspect datasets, train predictive models, measure fairness across demographic groups, apply bias mitigation techniques, and generate comprehensive audit reports. Through the integration of fairness metrics, explainability methods, and mitigation strategies, TrustLens aims to support the development of transparent and equitable AI systems.

The primary objective of this project is to build a practical platform that helps identify unfair treatment in machine learning predictions and evaluate the effectiveness of different fairness interventions. By combining model performance analysis with fairness evaluation, the system provides valuable insights into the trade-offs between predictive accuracy and equitable outcomes.

2. Project Overview

TrustLens is a web-based fairness auditing platform developed using Python and Streamlit. The system enables users to load real-world datasets, train machine learning models, evaluate fairness metrics, and generate detailed reports summarizing the findings. The platform supports multiple datasets, including the Adult Census Income dataset, IBM HR Analytics dataset, and a synthetic hiring dataset designed specifically for fairness experimentation.

The framework follows a structured workflow beginning with dataset analysis and preprocessing. After data preparation, multiple classification algorithms are trained and evaluated. Fairness metrics are then calculated to determine whether different demographic groups receive comparable outcomes. If disparities are detected, users can apply bias mitigation techniques and observe their impact on both fairness and predictive performance. To improve transparency, SHAP-based explainability methods are incorporated to highlight the influence of individual features on model predictions.

3. Objectives

- To develop an end-to-end fairness auditing framework for machine learning models.
- To identify and analyze demographic bias present in tabular datasets and predictive models.
- To evaluate model fairness using established metrics such as Statistical Parity Difference (SPD), Disparate Impact Ratio (DIR), Equal Opportunity Difference (EOD), and Equalized Odds Difference.
- To compare the performance of multiple machine learning algorithms, including Logistic Regression, Random Forest, and XGBoost.
- To detect unfair outcomes across protected groups such as gender, race, marital status, and age groups.
- To implement and assess bias mitigation techniques at the pre-processing, in-processing, and post-processing stages.
- To provide explainable AI capabilities using SHAP for understanding model decisions and feature influence.
- To analyze the trade-off between predictive performance and fairness after applying mitigation techniques.
- To generate automated fairness audit reports with actionable insights and recommendations.
- To support Responsible AI practices by promoting transparency, accountability, and fairness in machine learning systems.

4. Methodology

The methodology adopted in TrustLens consists of four major stages: data preparation, model training, fairness evaluation, and bias mitigation. Initially, the selected dataset is processed through a preprocessing pipeline that handles missing values, encodes categorical variables, standardizes numerical features, and creates training and testing partitions.

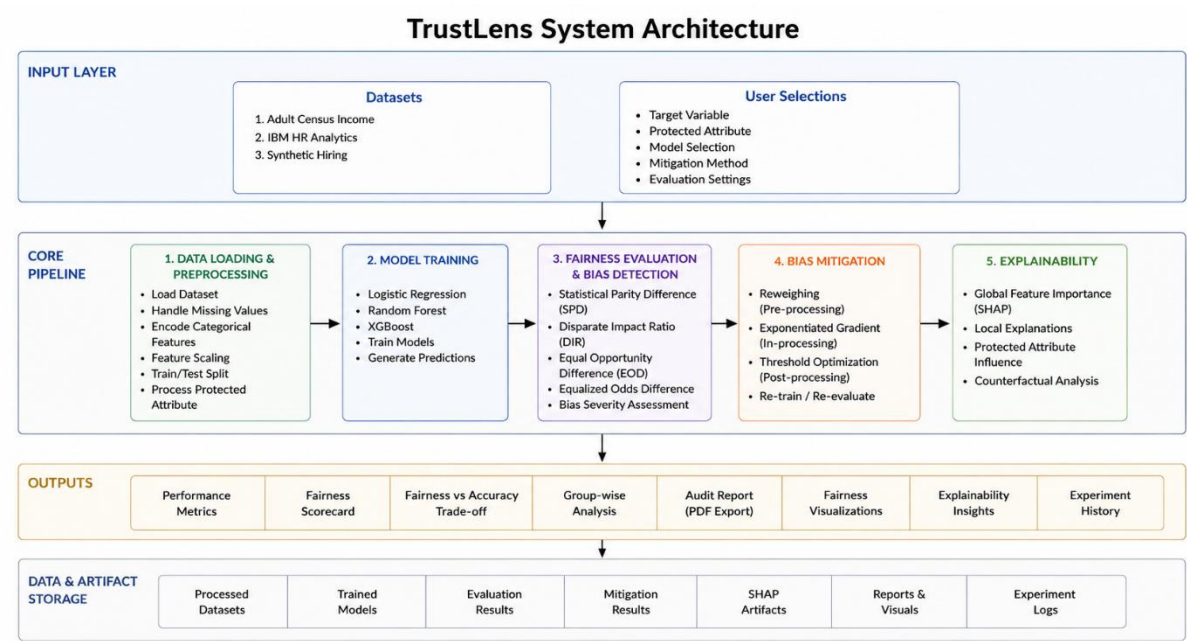
Three classification algorithms were selected for evaluation: Logistic Regression, Random Forest, and XGBoost. Logistic Regression serves as a simple and interpretable baseline model, while Random Forest and XGBoost provide more advanced ensemble-based approaches capable of capturing complex relationships within the data. These models are trained using the processed dataset and evaluated using standard performance measures such as accuracy, precision, recall, F1-score, and ROC-AUC.

To assess fairness, the project employs four widely recognized fairness metrics: Statistical Parity Difference (SPD), Disparate Impact Ratio (DIR), Equal Opportunity Difference (EOD), and Equalized Odds Difference. These metrics provide complementary perspectives on demographic fairness by examining differences in selection rates, positive outcome ratios, and classification error rates across protected groups.

When bias is detected, the framework offers multiple mitigation approaches. Reweighting is used as a preprocessing technique to balance group representation within the training data. Fairlearn's Exponentiated Gradient algorithm is applied as an in-processing approach that incorporates fairness constraints directly into model training. Additionally, Threshold

Optimization is used as a post-processing method that adjusts decision thresholds to improve fairness after training has been completed.

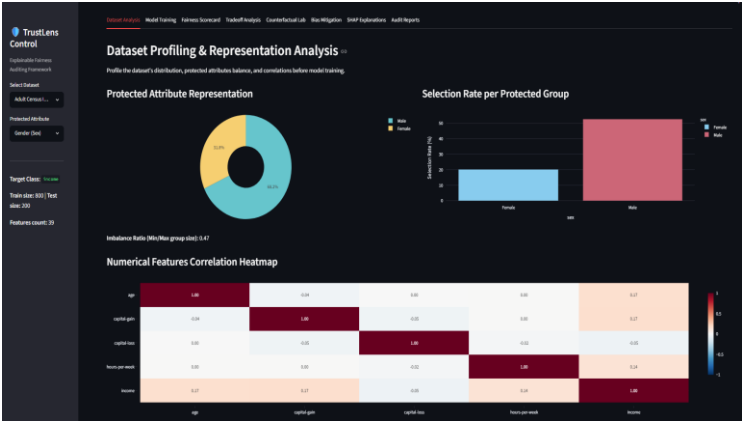
5. System Architecture

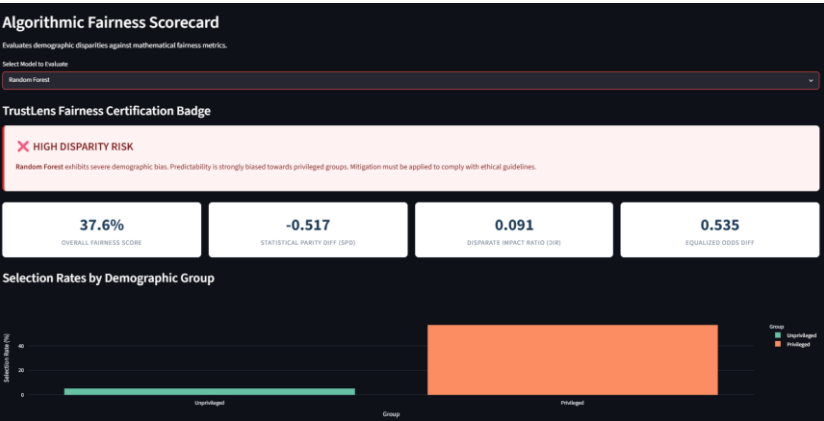
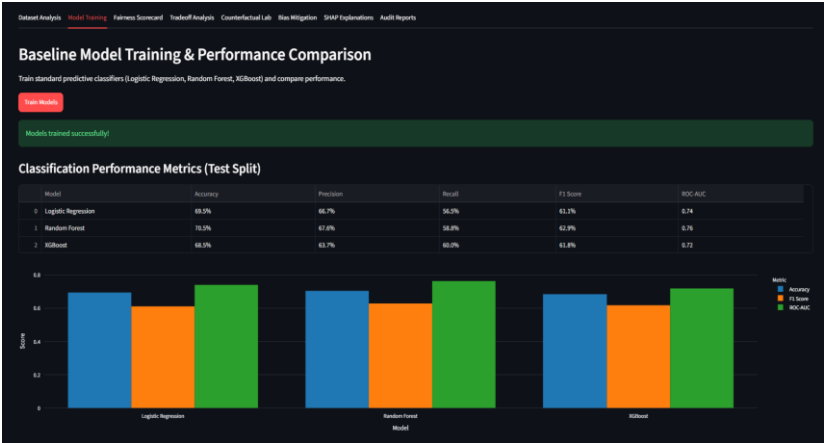


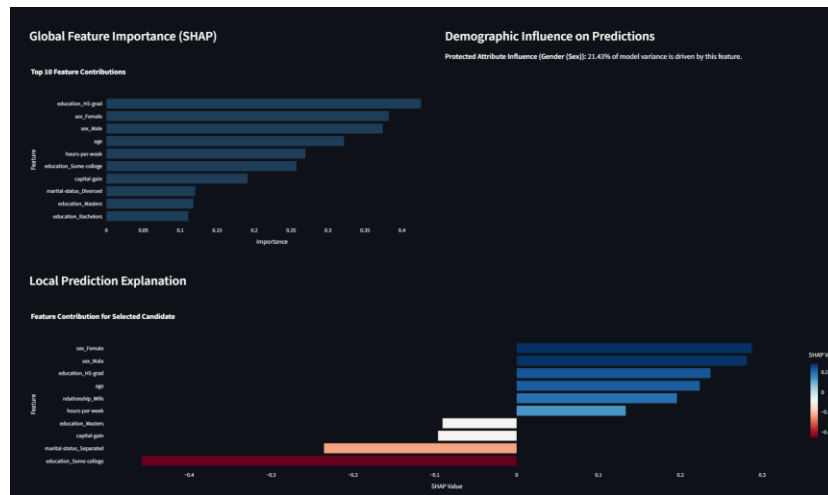
The architecture of TrustLens is organized into modular components that collectively support the fairness auditing process. The dataset loader module is responsible for importing and preprocessing data. The model training module manages classifier training and evaluation. Fairness assessment is handled through dedicated metric computation and bias detection modules, while explainability is provided through SHAP-based interpretation tools. Finally, a reporting module compiles the results into a structured PDF report.

This modular design improves maintainability and extensibility by allowing individual components to be modified or enhanced independently. Furthermore, the architecture enables users to compare different datasets, models, and mitigation techniques through a unified interface.

6. Results and Observations







Audit Report Generation & Insights Summary

Review dynamic findings generated by the Research Findings Engine, and download a detailed PDF report.

TrustLens Automatic Research Findings

- Random Forest achieved the highest baseline predictive accuracy (70.5%) on the test set.
- Random Forest exhibited the largest baseline demographic disparity, with a Statistical Parity Difference (SPD) of -0.517 (Disparate Impact Ratio of 0.091).
- Mitigation using Reweighing on Logistic Regression successfully reduced Statistical Parity Difference by 88.9% (down to SPD of 0.056).
- The fairest configuration was Logistic Regression with Reweighing mitigation, achieving an Overall Fairness Score of 93.2% while maintaining 61.5% accuracy.

Generate Audit Report PDF

Audited Model Configuration

Logistic Regression

☐ Include Mitigated Configuration in PDF

[Generate Audit Report](#)

The experimental results demonstrate the importance of evaluating fairness alongside traditional performance metrics. While baseline machine learning models often achieve strong predictive accuracy, fairness analysis reveals measurable disparities between demographic groups. Metrics such as Statistical Parity Difference and Disparate Impact Ratio help identify unequal outcome distributions that may not be evident from accuracy scores alone.

The application of bias mitigation techniques generally reduces demographic disparities, although slight reductions in predictive performance may occur. This observation highlights the inherent trade-off between fairness and accuracy in machine learning systems. The fairness scorecard and visualization dashboards implemented within TrustLens provide a practical means of understanding these trade-offs and selecting mitigation strategies appropriate for specific use cases.

The integration of SHAP explainability further enhances transparency by identifying the features that contribute most significantly to model predictions. These explanations enable users to investigate whether sensitive attributes or correlated variables exert disproportionate influence on decision outcomes.

7. Conclusion

TrustLens successfully demonstrates a complete fairness auditing workflow for tabular machine learning systems. The framework integrates dataset analysis, model training, fairness evaluation, bias mitigation, explainability, and report generation within a single platform.

Through the use of multiple fairness metrics and mitigation techniques, the system provides meaningful insights into demographic disparities and supports the development of more equitable machine learning models.

The project emphasizes that fairness should not be treated as a secondary consideration but rather as a fundamental component of responsible AI development. By combining technical evaluation with explainable insights, TrustLens offers a practical solution for organizations seeking to assess and improve the fairness of their machine learning systems.

8. References

Datasets Used

1. Adult Census Dataset
 - <https://www.kaggle.com/datasets/uciml/adult-census-income>
2. IBM HR Analytics Employee Attrition Dataset
 - <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>
3. hiring_dataset.csv (Synthetic Hiring dataset)

Code and implementation link:

<https://github.com/euph0x1/TrustLens-ML-Bias-Detection>